

Answer Key — Biology Grade 10

Sub-Strand 1.2: Chemicals of Life • Lesson 4: Proteins and Lipids: Do They Cause the Glucose Spike?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	B	C	A	C	D	B

USE AFTER: PREDICT PHASE — Predict: teacher asks "Which test method from Lesson 3 could we adapt to check?"

Q1. Before the tests, a learner says: "Iodine shows starch in ugali, so I will use iodine to find out whether beans contain protein." Which response is correct?

- A. Iodine will turn violet if protein is present, so the plan works
- B. Iodine tests for all food chemicals, so it will show protein and fat as well
- C. Iodine tests for starch only, so a different test (the biuret test) is needed for protein**
- D. Protein cannot be tested for in the laboratory, so beans must be judged by taste

Correct answer: C. Iodine reacts with starch (blue-black). Protein needs its own test, the biuret test, which uses sodium hydroxide and dilute copper sulphate.

USE AFTER: OBSERVE PHASE — Test A: biuret test for protein on beans, milk, nyama choma and ugali

Q2. A group adds 2 cm³ of dilute sodium hydroxide and then 2-3 drops of dilute copper sulphate to a tube of bean paste mixed with water. The mixture turns violet (purple). What does this result show?

- A. The beans contain starch
- B. The beans contain protein**
- C. The beans contain lipids (fats and oils)
- D. The beans contain reducing sugar

Correct answer: B. A blue to violet/purple colour change in the biuret test shows that protein is present. Starch gives blue-black with iodine, reducing sugar gives a brick-red colour with Benedict's, and lipids give a translucent spot or cloudy emulsion.

USE AFTER: OBSERVE PHASE — Test B (i): grease-spot test on groundnuts, mandazi, milk and ugali

Q3. A learner rubs crushed groundnuts on a piece of brown paper, dries it gently and holds it up to the light. A translucent (see-through) spot is left on the paper. What does this show?

- A. The groundnuts contain starch
- B. The groundnuts contain protein
- C. The groundnuts contain lipids (oil)**
- D. The groundnuts contain only water, which has made the paper thin

Correct answer: C. Lipids leave a translucent grease spot that does not disappear when the paper dries. Water would evaporate on drying, and the protein and starch tests use different reagents.

USE AFTER: EXPLAIN PHASE — Explain: comparison table of carbohydrates, proteins and lipids (building blocks)

Q4. Which row correctly matches each food chemical to the smaller units it is built from?

- A. Carbohydrates: glucose units; proteins: amino acids; lipids: fatty acids and glycerol**
- B. Carbohydrates: glucose units; proteins: fatty acids and glycerol; lipids: amino acids
- C. Carbohydrates: amino acids; proteins: glucose units; lipids: fatty acids and glycerol
- D. Carbohydrates: fatty acids and glycerol; proteins: glucose units; lipids: amino acids

Correct answer: A. Starch and other carbohydrates are chains of glucose units, proteins are chains of amino acids, and lipids are made of fatty acids and glycerol.

USE AFTER: EXPLAIN PHASE — Explain: Claim-Evidence-Reasoning using the meal numbers (70 g carbohydrate, 10 g fat)

Q5. Wanjiru's ugali contains about 70 g of carbohydrate and about 10 g of fat. About how many times more carbohydrate than fat is this?

- A. 60 times
- B. 80 times
- C. 7 times**
- D. 0.14 times

Correct answer: C. $70\text{ g} \div 10\text{ g} = 7$, so there is about 7 times more carbohydrate than fat. This supports the claim that carbohydrate, not fat, is the main cause of the glucose rise from 4.5 to 8.0 mmol/L.

USE AFTER: EXPLAIN PHASE — Explain: "Why do beans, milk and groundnuts still matter for Wanjiru...?"

Q6. Beans, milk and groundnuts are not the main cause of the glucose spike after a meal. Why do they still matter for the body?

- A. They are turned into glucose immediately, so they cause the spike as well
- B. They only add sweetness, so they have no use in the body
- C. They replace carbohydrates completely, so ugali is not needed
- D. Proteins in them are used for growth and repair of body tissues, and lipids are used for energy storage**

Correct answer: D. Proteins are needed for growth and repair of body tissues, and lipids store energy and give insulation. Their roles differ from carbohydrates, which are the main energy source and cause the spike.

USE AFTER: MODEL BUILDING PHASE — Model: expanding the 'Food chemicals' box with thick and thin arrows

Q7. Learners are updating their flow-diagram model of what happens to a meal of ugali, beans and cooking oil. Which arrow should be drawn thick, as the main cause of the glucose spike in the blood?

- A. Proteins (beans, nyama choma) to glucose in blood
- B. Carbohydrates (starch in ugali) to glucose in blood**
- C. Proteins and lipids to glucose in blood, equally with carbohydrates
- D. Lipids (cooking oil, groundnuts) to glucose in blood

Correct answer: B. The meal data (about 70 g carbohydrate against about 20 g protein and 10 g fat) and the test results link carbohydrates to the glucose rise. Proteins and lipids go on thinner arrows towards growth, repair and energy storage.